

Introduction

Types of urban flooding differ between fluvial, pluvial and tidal, with varying impacts and resilience needs.¹ Identifying the type of flooding experienced by New York City neighborhoods can inform subsequent flood resilience policy. Project FloodNet provides frequent, publicly available flood observations across NYC to support this cause.

The initial purpose of this project was to classify FloodNet sensors as tidally influenced or non-tidally influenced, using the dataset published on NYC Open Data.²

During this process, a larger data quality issue emerged: several 2025 flood observations were either partially or completely missing from the dataset.

This two-part study first classifies sensors by tidal influence, then explores the potential data quality issues within the FloodNet dataset.

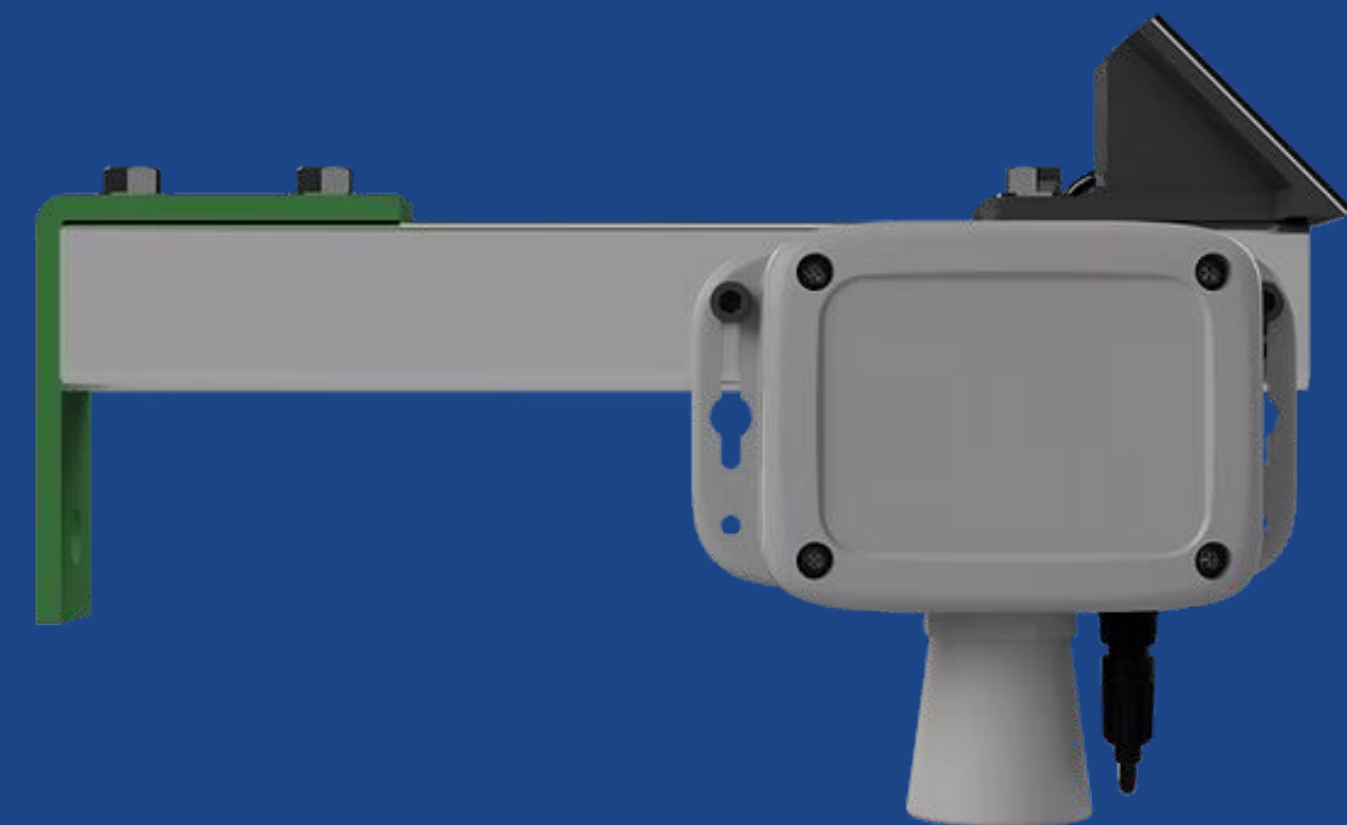


Image of a FloodNet Sensor.³ The sensor shoots an ultrasonic wave downwards, and measures the time it takes for the wave to reflect and return. This measurement is then used to calculate height above ground level, indicating a flood if $h > 0$.

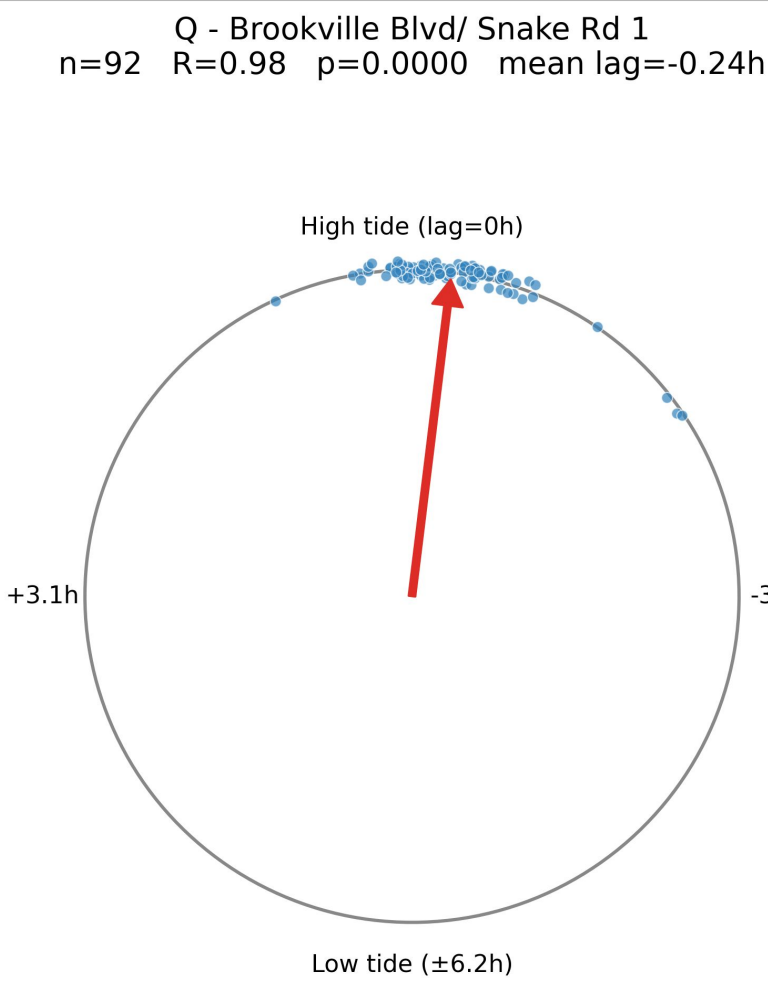
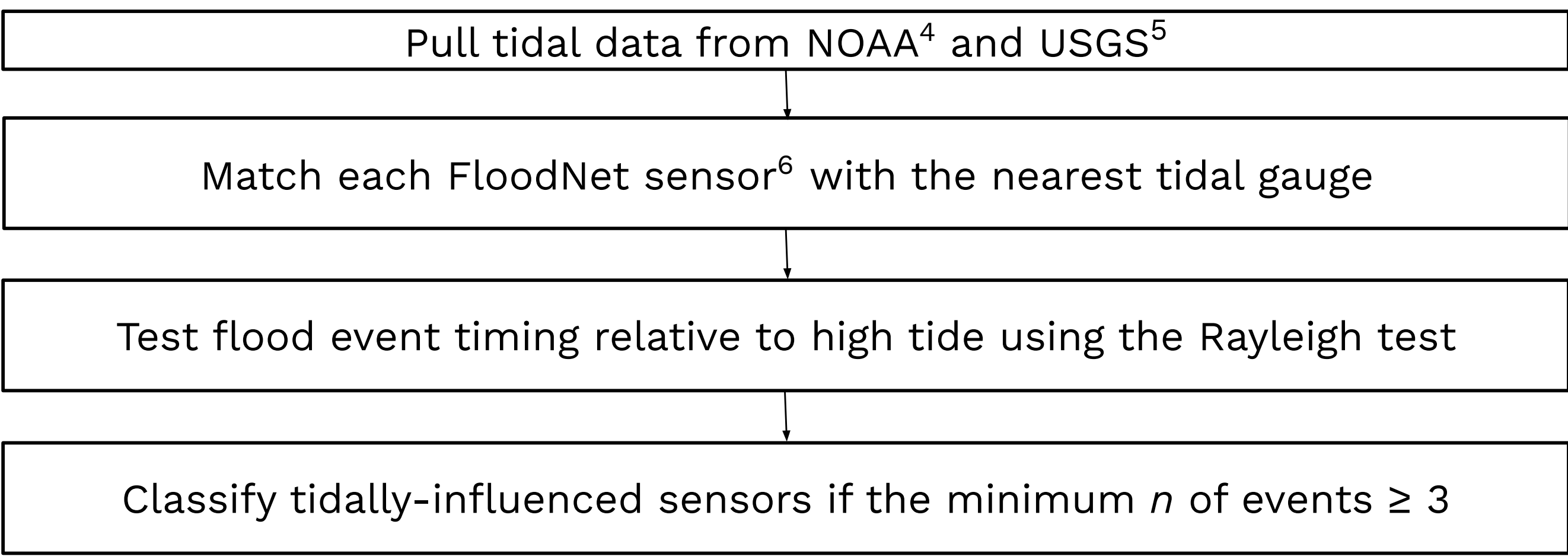
Objectives & Hypotheses

This study involves two key objectives:

- 1. Separate tidally-influenced flood sensors from non-tidally-influenced sensors.**
 - Hypothesis:* Coastal floodplain sensors will show stronger tidal influence than inland sensors.
- 2. Detect data quality gaps that might be wrongly-influencing downstream FloodNet analyses.**
 - Hypothesis:* No data quality gaps exist in the publicly-deployed FloodNet dataset.

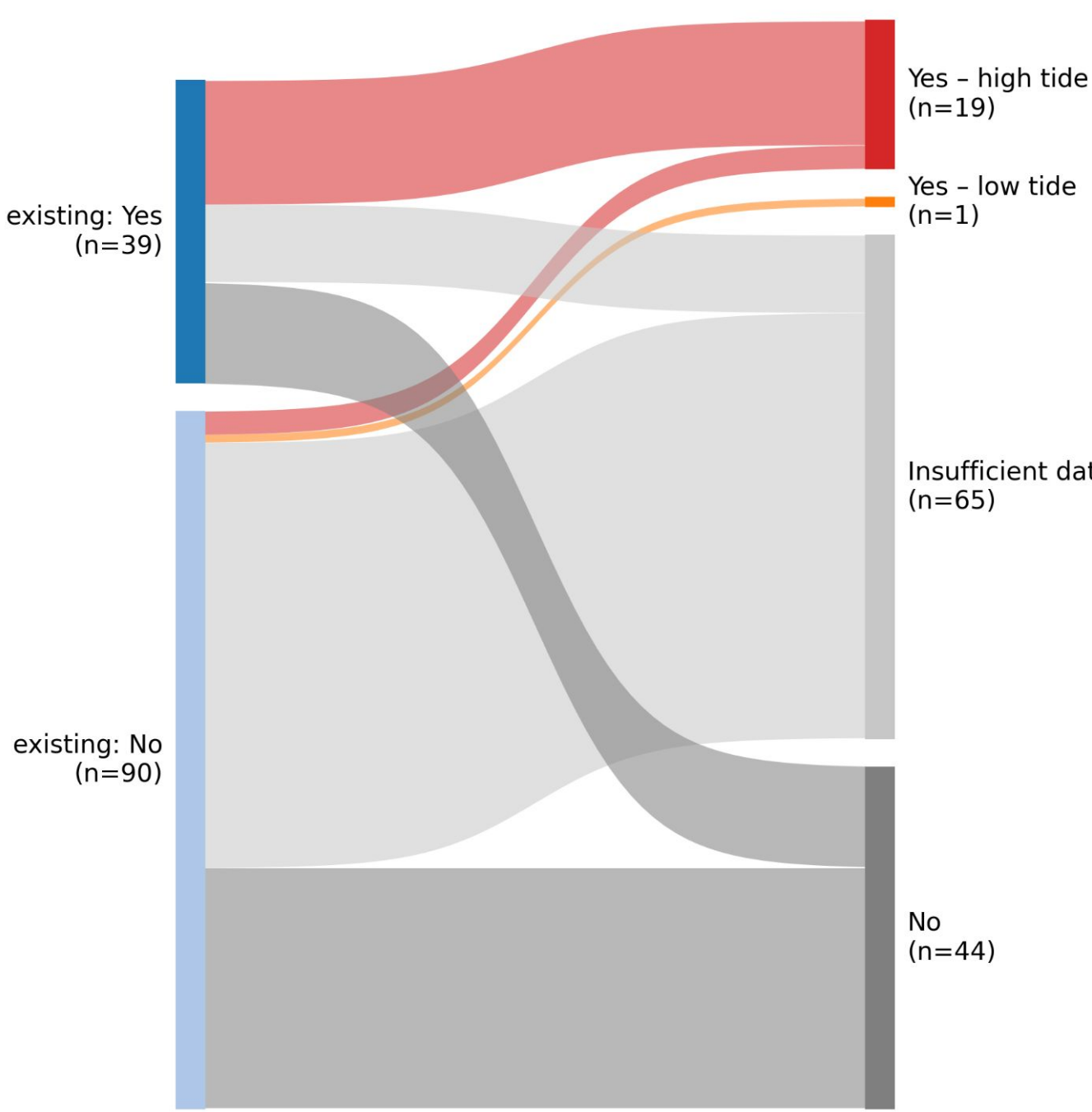
Part 1: The Tidal Classification Analysis

Methodology

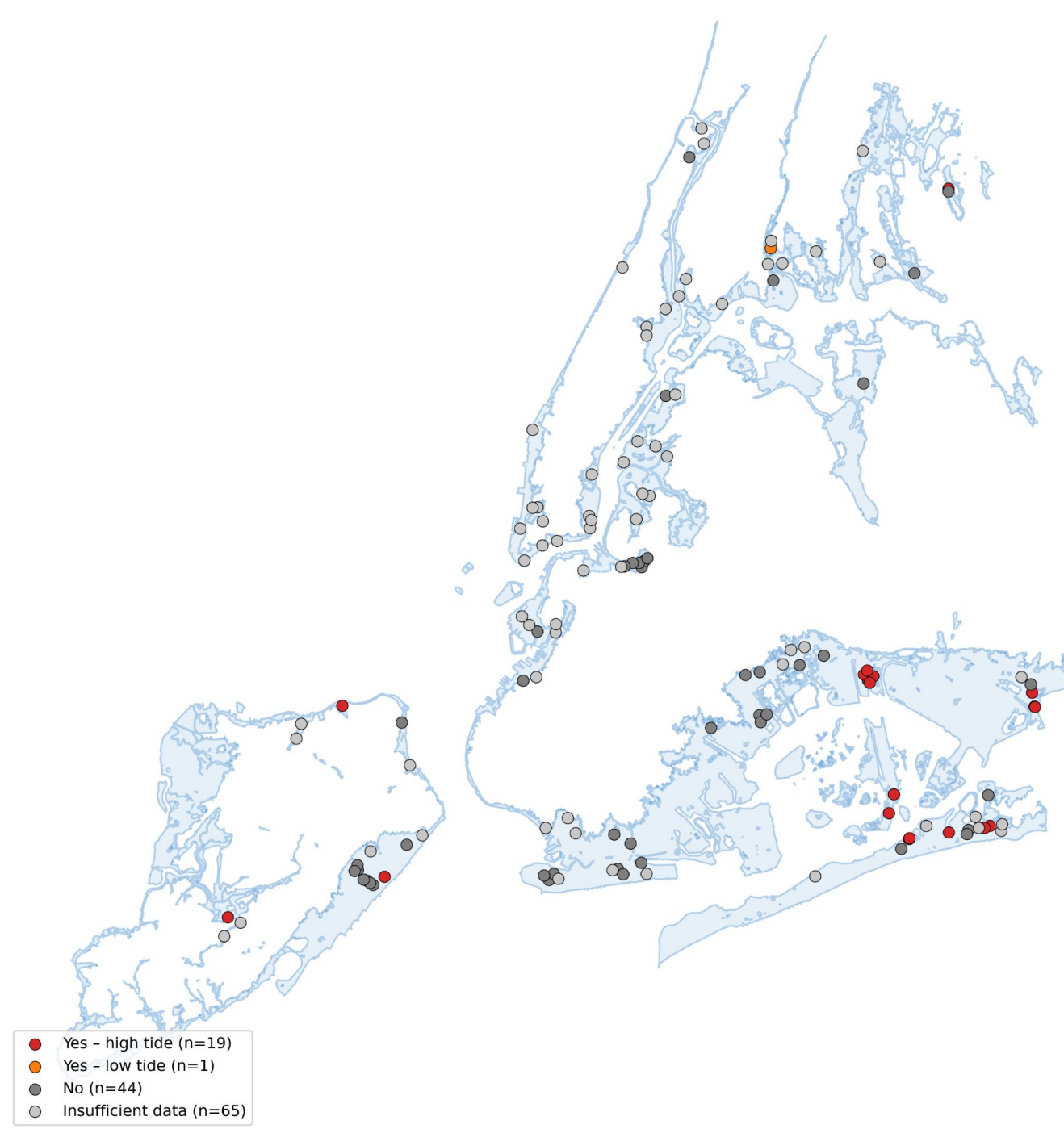


Results

Existing metadata vs. Rayleigh-test tidal classification



Sensor tidal classification (Rayleigh test on flood-event lag)



- **1 sensor** associated with **low tide**
- **19 sensors** associated with **high tide**
- **44 sensors** not associated with tides
- Excluding insufficient data, **17 sensors changed** their classification status

Key finding: 65 sensors indicate insufficient data available.

Question: Why are there so many cases of insufficient data? This warrants a second look at the data itself.

Conclusions

The tidal classification analysis conducted in Part 1 found that many sensors lacked sufficient data for a confident result, motivating a second investigation into FloodNet's publicly-available data quality. Cross-checking sensor records with rainfall, high tides, and 311 flooding reports revealed data gaps in 2025 and identified potential missing flood events.

A larger study would help determine how many sensors are affected by this data quality issue and whether the gap extends beyond 2025.

Future work will evaluate sensor activity and frequency to inform optimal placements for FloodNet sensors across New York City.

Part 2: The Data Quality Check

Methodology



Manually log missing events from 2025 for sensors that switched tidal classifications

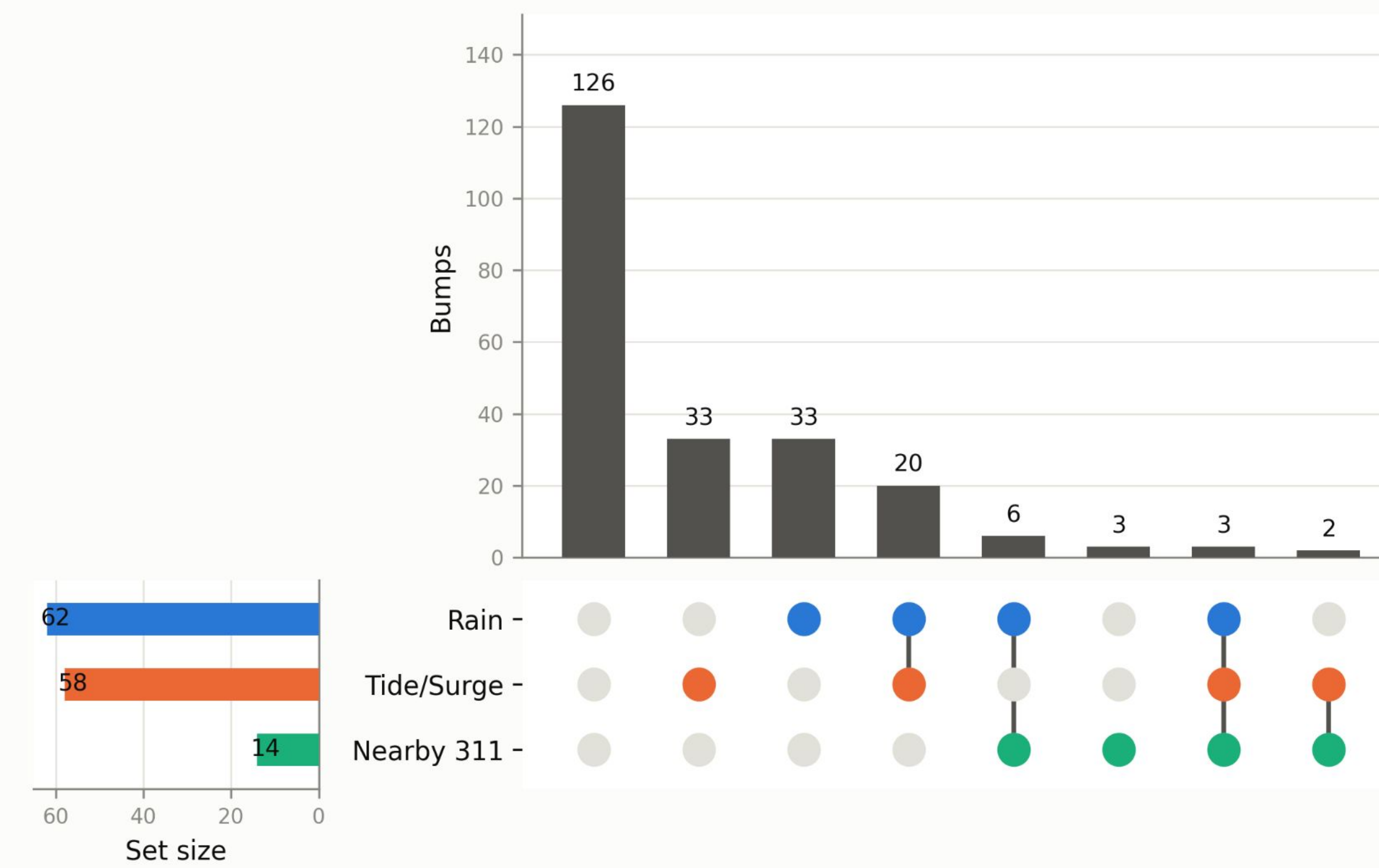
Match nearest rain⁷, tide/storm surge, and 311 call signal⁸ for each missing event date

Check for rain, tide/surge, or 311 flooding signal on event day (± 1 day)

Classify each event as corroborated (real flood, data gap) vs. uncorroborated (weak/no evidence)

Results

Corroborating signals for 226 manually-flagged FloodNet bumps (2025)



- **56%** (126 bumps) have **0 corroborating elements**.
- **30.5%** (69 bumps) have **1 corroborating element**.
- **12.4%** (28 bumps) have **2 corroborating elements**.
- **1.3%** (3 bumps) have **all 3 corroborating elements**.

Key findings: Nearly **14% of sensor-detected events** that went unreported have *at least* 2 corroborating elements of flood-related evidence.

References

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